NAME: Aryan Singh STD.: _____ SEC.: _____ ROLL NO.: _____ SUB.: _____

S. No.	Date	Title	Page No.	Teacher's Sign / Remarks
		Index		
2	12-Aug-25	full adder and subtractor verilog code		12/8/25
3	26-Aug-25	Decoder using test Bench		26/8/25
4	26-Aug-25	Multiplexer And Demultiplexer Code in verilog		26/8/25

(*) Aim :- To design stimulate and verify the magnification functionality of various fundamental combinational logic circuits using verilog HDL.

(*) Theory :-

This experiment covers the Design And verification of standard combinational circuits and fundamental concepts of the verilog language.

(*) Tools required :-

(1) Model Sim :- This is an HDL simulation tool used for verifying the functionality of digital circuits.

(2) Verilog HDL :- This is the language used to describe the digital circuits and their best benches.

Date
12-Aug-25

{Assesment - 2}

Full Adder

```
module full_adder (a, b, cin, sum, carry, d, bo);
input a, b, cin;
output sum, carry;
output d, bo;
assign sum = a ^ b ^ cin;
assign carry = (a & b) | cin & (a ^ b);
assign d = a ^ b ^ cin;
assign bo = ((~a) & b) | ((a ^ b) & (~cin));
endmodule
module tb();
reg a, b, cin;
wire sum, carry;
wire d, bo;
full_adder m0 (a, b, cin, sum, carry, d, bo);
initial
begin
a = 1'b0; b = 1'b0; cin = 1'b0;
#100 a = 1'b0; b = 1'b0; cin = 1'b0;
#100 a = 1'b0; b = 1'b0; cin = 1'b0;
#100 a = 1'b0; b = 1'b0; cin = 1'b0;
#100 a = 1'b0; b = 1'b0; cin = 1'b0;
#100 a = 1'b0; b = 1'b0; cin = 1'b0;
#100 a = 1'b0; b = 1'b0; cin = 1'b0;
#100 a = 1'b0; b = 1'b0; cin = 1'b0;
#100 a = 1'b0; b = 1'b0; cin = 1'b0;
end
endmodule
```

O/P verified
12/8/25
24BCE0403

```
module Full_Adder(a,b,cin,sum,carry,d,b0);
input a,b,cin;
output sum,carry;
output d,b0;
assign sum=a^b^cin;
assign carry=(a&b)|cin&(a^b);
assign d=a^b^cin;
assign b0=((~a)&b)|((a^b)&(~cin));
endmodule
module tb();
reg a,b,cin;
wire sum,carry;
wire d,b0;
Full_Adder mo(a,b,cin,sum,carry,d,b0);
initial
begin
a=1'b0;b=1'b0;cin=1'b0;
#100
a=1'b0;b=1'b0;cin=1'b0;
#100
a=1'b0;b=1'b0;cin=1'b0;
#100
a=1'b0;b=1'b0;cin=1'b0;
#100
a=1'b0;b=1'b0;cin=1'b0;
#100
a=1'b0;b=1'b0;cin=1'b0;
#100
a=1'b0;b=1'b0;cin=1'b0;
#100
a=1'b0;b=1'b0;cin=1'b0;
end
endmodule
```

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ModelSim - INTEL FPGA STARTER EDITION 2020.1

late Add Wave Tools Layout Bookmarks Window Help

Layout simulate ColumnLayout AllColumns

100 ns

n unit	Design unit type	Top Category	Visibility	Tot
	Module	DU Instance	+acc=<f...	
dder	Module	DU Instance	+acc=<f...	
	Capacity	Statistics	+acc=<f...	

/home/matlab/24BCT0194_Thomas/full adder and substractor.v (/tb) - Default

226 ps

```
Ln#
1  module FullAdder(a,b,cin,sum,carry,d,b0);
2   input a,b,cin;
3   output sum,carry;
4   output d,b0;
5   assign sum=a^b^cin;
6   assign carry=(a&b)|cin&(a^b);
7   assign d=a^b^cin;
8   assign b0=(~a&b)|(cin & (~(a^b)));
9   endmodule
10  module tb();
11  reg a,b,cin;
12  wire sum,carry;
13  wire d,b0;
14  FullAdder mo(a,b,cin,sum,carry,d,b0);
15  initial
16  begin
17  a=1'b0;b=1'b0;cin=1'b0;
18  #100a=1'b0;b=1'b0;cin=1'b0;
19  #100a=1'b0;b=1'b0;cin=1'b1;
20  #100a=1'b0;b=1'b1;cin=1'b0;
21  #100a=1'b0;b=1'b1;cin=1'b1;
22  #100a=1'b1;b=1'b0;cin=1'b0;
23  #100a=1'b1;b=1'b0;cin=1'b1;
24  #100a=1'b1;b=1'b1;cin=1'b0;
25  #100a=1'b1;b=1'b1;cin=1'b1;
26  end
27  endmodule
28
```

Wave - Default

Msgs	
/tb/a	0
/tb/b	0
/tb/cin	1
/tb/sum	St1
/tb/carry	St0
/tb/d	St1
/tb/b0	St1

Now 100000 ps

Cursor 1 226 ps

ps 200 ps 400 ps 600 ps

226 ps

er
ertpoint sim:/tb/*

Project : baladitya Now: 100 ns Delta: 0 /tb/sum

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Date
Page

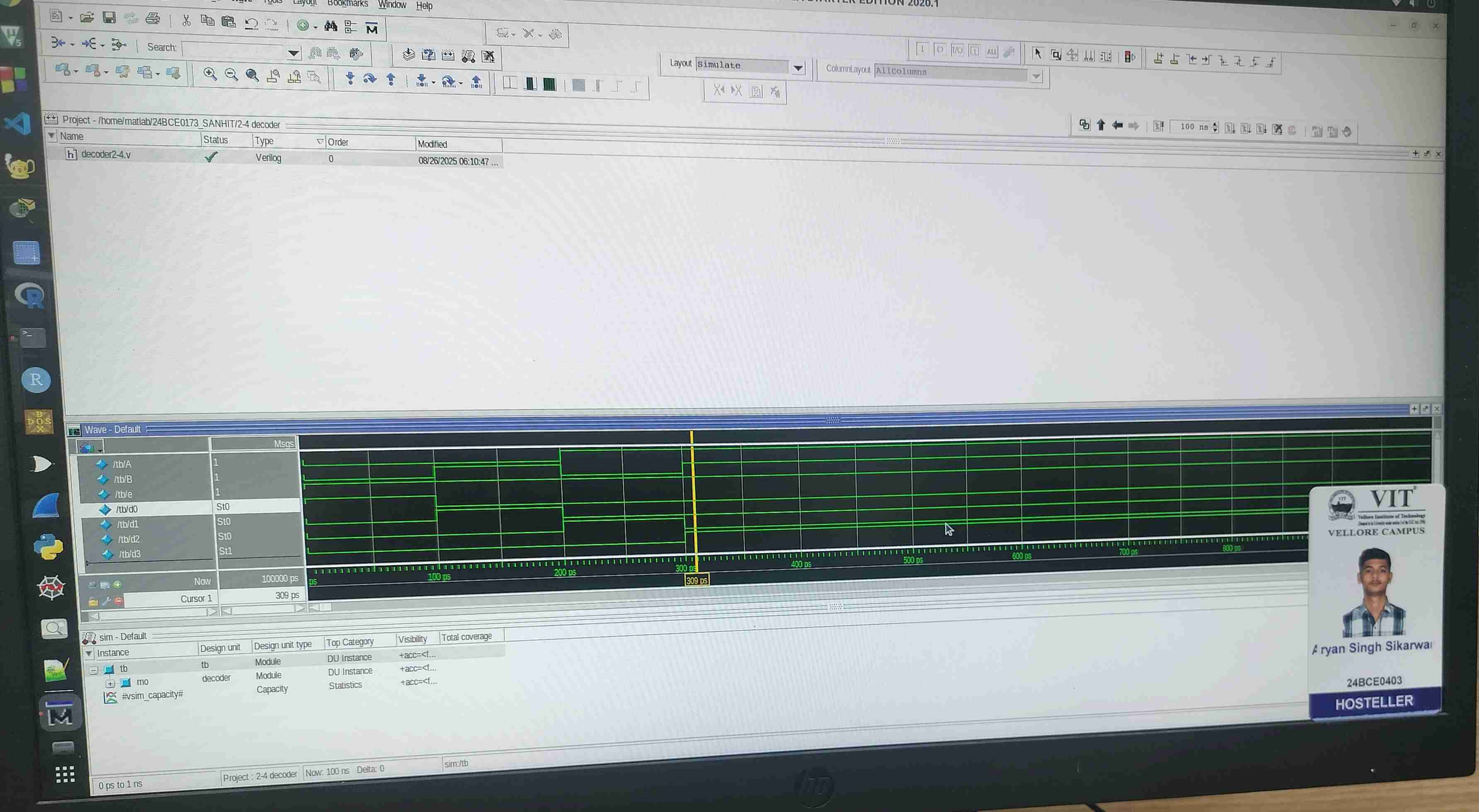
T → 0
F = 1

A	B	C	D	Bo	
T	T	T	T	T	
T	T	F	F	F	
T	F	T	F	F	
T	F	F	T	F	
F	T	T	F	T	
F	T	F	T	T	
F	F	T	T	T	
F	F	F	F	F	

#2 Decoder Test Bench code :-

```
module decoder (A, B, e, d0, d1, d2, d3);
input A, B;
output d0, d1, d2, d3;
assign d0 = ~A & ~B & e;
assign d1 = ~A & B & e;
assign d2 = A & ~B & e;
assign d3 = A & B & e;
endmodule tb();
reg A, B, e;
wire d0, d1, d2, d3;
decoder dec (A, B, e, d0, d1, d2, d3);
Initial
begin
A=1'b0; B=1'b0; e=1'b1;
#100 A=1'b0; B=1'b1; e=1'b1;
#100 A=1'b1; B=1'b0; e=1'b1;
#100 A=1'b1; B=1'b1; e=1'b1;
end
endmodule
```

o/p verified
26/8/25
24BEC04003


Project - /home/matlab/24BCE0173_SANHIT/2-4 decoder
Name
Status
Type
Order
Modified
decoder2-4.v
Verilog
0
08/26/2025 06:10:47 ...
Wave - Default
Msgs
/tb/A
/tb/B
/tb/e
/tb/d0
/tb/d1
/tb/d2
/tb/d3
Now
100000 ps
Cursor 1
309 ps
sim - Default
Instance
Design unit
Design unit type
Top Category
Visibility
Total coverage
tb
mo
#vsim_capacity#
decoder
Module
Capacity
DU Instance
Statistics
+acc=<f...
0 ps to 1 ns
Project : 2-4 decoder
Now: 100 ns Delta: 0
sim:/tb
VIT
VELLORE CAMPUS
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24BCE0403
HOSTELLER

Multiplexer Code :-

```
module Multiplexer (d0, d1, d2, d3, d4, d5
, d7, sel, out) ;
input d0, d1, d2, d3, d4, d5, d6, d7;
input [2:0] sel ;
output reg out ;
always @ (sel) :
begin
case (sel) :
3'b000 : out d0 ;
3'b001 : out d1 ;
3'b010 : out d2 ;
3'b011 : out d3 ;
3'b100 : out d4 ;
3'b101 : out d5 ;
3'b110 : out d6 ;
3'b111 : out d7 ;
end
endmodule
```

```
module Testbench ;
reg d0, d1, d2, d3, d4, d5, d6, d7;
reg [2:0] sel ;
wire out ;
multipuxer out (.d0(d0), .d1(d1), .d2(d2)
, .d3(d3), .d4(d4) , .d5(d5), .d6(d6)
, .d7(d7), .sel(sel) , .(out) out));
initial begin
d0=0 ; d1=0 ; d2=0 ; d3=0 ; d4=0;
d5=0 ; d6=0 ; d7=0 ; sel=0;
# 100 ;
d0=0 ; d1=0 ; d2=0 ; d3=0;
d4=1 ; d5=1 ; d6=0 ; d7=1;
```

24BCE0403

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/home/matlab/Arya_24bce2761/2532.v (/Testbench)

File Edit View Tools Bookmarks Window Help

/home/matlab/Arya_24bce2761/2532.v (/Testbench) - Default

```
Ln#
 1  module Multiplexer(d0,d1,d2,d3,d4,d5,d6,d7,sel,out);
 2  input d0,d1,d2,d3,d4,d5,d6,d7;
 3  input[2:0]sel;
 4  output reg out;
 5  always @(sel)
 6  begin
 7  case(sel)
 8  3'b000:out=d0;
 9  3'b001:out=d1;
10  3'b010:out=d2;
11  3'b011:out=d3;
12  3'b100:out=d4;
13  3'b101:out=d5;
14  3'b110:out=d6;
15  3'b111:out=d7;
16  endcase
17  end
18  endmodule
19
```

80 ps

Ln: 1 Col: 0

Wave - Default

Msgs	
/Testbench/d0	0
/Testbench/d1	0
/Testbench/d2	0
/Testbench/d3	0
/Testbench/d4	0
/Testbench/d5	0
/Testbench/d6	0
/Testbench/d7	0
/Testbench/sel	000
/Testbench/out	St0

000 101

200 ps 400 ps 600 ps 800 ps 1000 ps

Now 1000 ps

Cursor 1 80 ps

80 ps

Project : 24bce2532 Now: 1 us Delta: 0 /Testbench/d6

Demultiplexer code :-

```
module demultiplexer (a,x,y,z,d0,d1,d2,
  d3,d4,d5,d6,d7);
input a,x,y,z;
output d0,d1,d2,d3,d4,d5,d6,d7;
assign d0 = (a&~x&~y&~z);
d1 = (a&~x&~y&z);
d2 = (a&~x&y&~z);
d3 = (a&~x&y&z);
d4 = (a&x&y&~z);
d5 = (a&x&y&z);
d6 = (a&x&y&~z);
d7 = (a&x&y&z);
endmodule

module Testbench_demux;
reg a,x,y,z;
wire d0,d1;d2,d3,d4 id5,d6,d7;
demultiplexer out(.a(a),.x(x),.y(y),
.z(z),.d0(d0),.d1(d1),.d2(d2),.d3(d3),
.d4(d4),.d5(d5),.d6(d6),.d7(d7));
initial
begin
a=0; x=0; y=0; z=0;
#100 a=1; x=0; y=1; z=0;
#100 a=1; x=1; y=0; z=0;
#100 a=1; x=1; y=1; z=1;
end
endmodule
```

Verified

26/08/25

24BCE0403

Activities WindowObj Aug 26 6:49 PM

ModelSim - INTEL FPGA STARTER EDITION 2020.1

File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help

sim - Default

Instance	Design unit	Design unit type	Top Category	Visibility	Total coverage
tb	tb	Module	DU Instance	+acc=<f...	
uut	de_mux	Module	DU Instance	+acc=<f...	
#vsim_capacity#		Capacity	Statistics	+acc=<f...	

Project sim

/home/matlab/zayn_mux.v (/tb) - Default

```
1  module de_mux(a,x,y,z,d0,d1,d2,d3,d4,d5,d6,d7);
2  input a,x,y,z;
3  output d0,d1,d2,d3,d4,d5,d6,d7;
4  assign d0 = (a& ~x& ~y&~z),
5  d1 = (a&~x&~y&z),
6  d2 = (a&~x&y&~z),
7  d3 = (a&~x&y&z),
8  d4 = (a&x&y&~z),
9  d5 = (a&x&y&z),
10 d6 = (a&x&y&~z),
11 d7 = (a&x&y&z);
12 endmodule
13 module tb;
14 reg a,x,y,z;
15 wire d0,d1,d2,d3,d4,d5,d6,d7;
16 de_mux uut(.a(a),.x(x),.y(y),.z(z),.d0(d0),.d1(d1),.d2(d2),.d3(d3),.d4(d4),.d5(d5),.d6(d6),.d7(d7));
17 initial begin
18 a=0;x=0;y=0;z=0;
19 #100 a=1;x=0;y=1;z=0;
20 #100 a=1;x=1;y=0;z=0;
21 #100 a=1;x=1;y=1;z=1;
22 end
23 endmodule
24
```

97751 ps

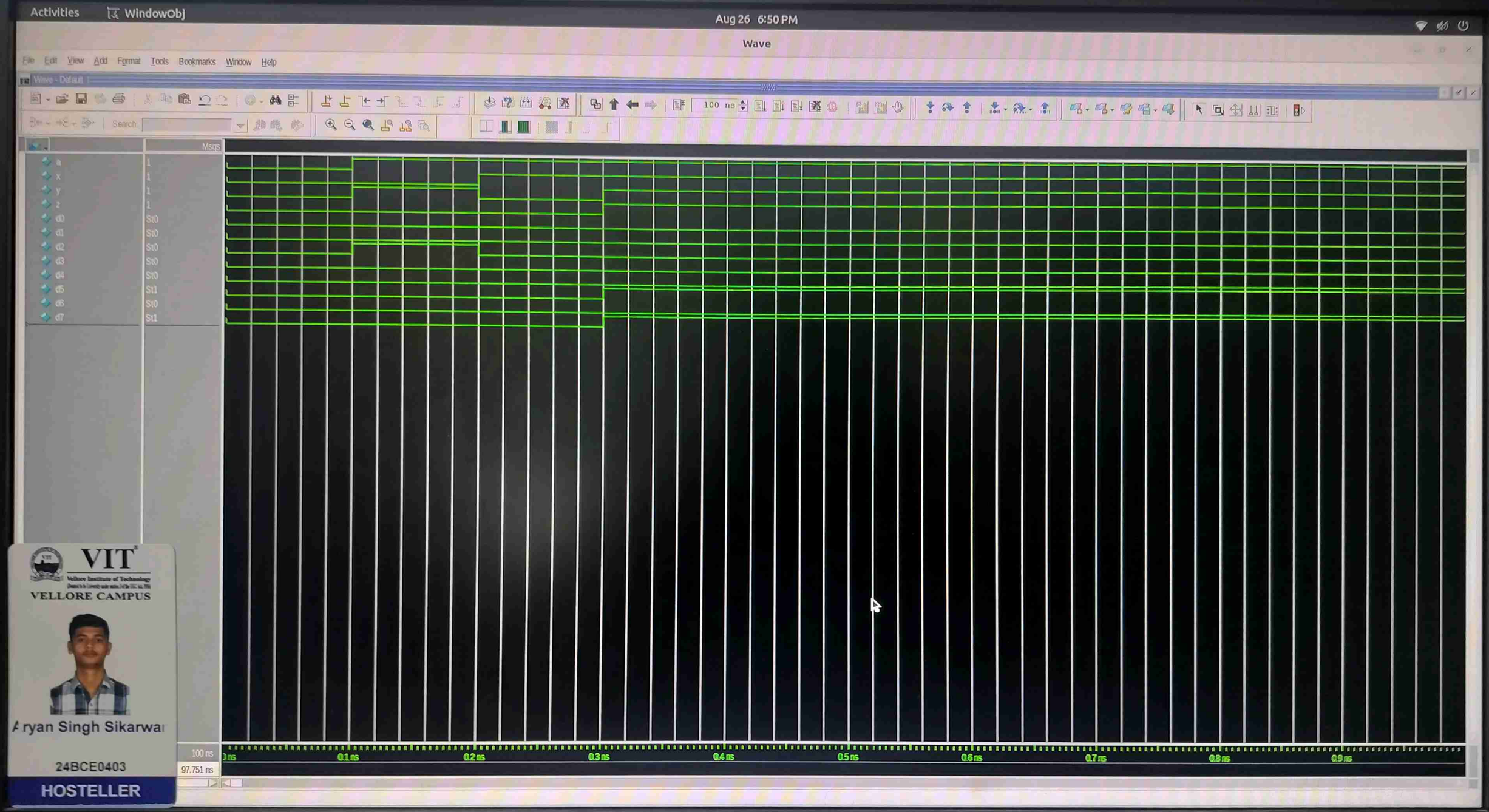
```
0 {50 ps} -r 100
d with 3 errors.
d with 2 errors.
d with 5 errors.
uccessful.
d with 5 errors.
uccessful.
uccessful.

6,2025, Elapsed time: 0:24:58

26,2025

sim:/tb/*
```

Now: 100 ns Delta: 0 sim:/tb


Activities
WindowObj
Aug 26 6:50 PM
Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Search:
Msgs
a
x
y
z
d0
d1
d2
d3
d4
d5
d6
d7
100 ns
0.1 ns
0.2 ns
0.3 ns
0.4 ns
0.5 ns
0.6 ns
0.7 ns
0.8 ns
0.9 ns
97.751 ns
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Layout Bookmarks Window Help

Layout NoDesign

Search

Top Category	Visibility	Total coverage
DU Instance	+acc=<f...	
DU Instance	+acc=<f...	
Statistics	+acc=<f...	

Search For

/home/matlab/debu.v (/tb) - Default

```
Ln#
 1  module decorder(A,B,e,d0,d1,d2,d3);
 2   input A,B,e;
 3   output d0,d1,d2,d3;
 4   assign d0 = ~A & ~B & e;
 5   assign d1 = ~A & B & e;
 6   assign d2 = A & ~B & e;
 7   assign d3 = A & B & e;
 8  endmodule
 9
10  module tb();
11   reg A,B,e;
12   wire d0,d1,d2,d3;
13   decorder dec(A,B,e,d0,d1,d2,d3);
14   initial
15  begin
16   A=1'b0;B=1'b0;e=1'b1;
17   #100 A = 1'b0;B = 1'b1;e=1'b1;
18   #100 A = 1'b1;B = 1'b0;e=1'b1;
19   #100 A = 1'b1;B = 1'b1;e=1'b1;
20  end
21  endmodule
22
```

Wave | debu.v

Project : dbeu | Now: 100 ns | Delta: 0 | sim:/tb

Answer the following

⊛ what do you mean by test Bench in verilog?

Ans It creates a copy of the module you want to test and places it inside the virtual lab. It seprates piece of a code you write to check if your main hardware design.

⊛ what is Difference between initial and always?

Ans The initial block runs only piece at the beginning of a simulation typically for setting up a test. The always block runs repeatedly and continuously, modeling the behaviour of real hardware always active.

⊛ what is Instalation in verilog?

Ans Instantiation is the process of creating a usable copy of one module inside another. Its the fundamental method for building compux, hierarchical circuits from smaller, reusable components.